Software Requirements & Architecture Document (SRAD)

Project: Multi-Tenant LMS & Dynamic Website Builder
Target Launch Market: Egypt (Phase 1) → MENA / GCC Region Expansion (Phase 2)
Business & Cost Strategy: $0 Initial Overhead Blueprint with Variable Transaction-Based Scaling

1. Core Technology Stack ($0 Infrastructure Strategy)

Component	Technology	Selection Rationale	Budget Tier
Frontend Framework	Next.js 15+ (App Router)	Essential for high-performance middleware routing and serverless dynamic generation.	Free (Vercel)
Styling & Components	Tailwind CSS + shadcn/ui	Allows client site styling and fast UI adjustments natively via global CSS variables.	Free
Database & Auth	Supabase (PostgreSQL)	Built-in relational multi-tenancy mapping, secure Row-Level Security, and Auth services.	Free Tier
Video Infrastructure	Cloudflare Stream	Secure streaming. Automatically scales on minutes stored/watched with zero monthly minimum fee.	Pay-as-you-go
Hosting & Core DNS	Vercel + Cloudflare DNS	Automated SSL generation and wild-card custom subdomain handling (*.platform.com).	Free Tier
Payment Rail (MVP)	Manual Verification Rails	Immediate activation. Eliminates commercial registration friction by verifying InstaPay screenshots.	$0 (No overhead)
Payment Rail (Scale)	Paymob / Tap Payments API	Direct integration with Vodafone Cash, Fawry kiosks, Meeza, and standard MENA credit systems.	Transaction-based

2. Functional User Stories

Epic 1: Instructor Onboarding & Site Customization

- **Story 1: Fast Account Creation** – As an instructor, I want to create an account in 60 seconds and reserve a custom subdomain (e.g., `physics-pro.platform.com`) to solidify my digital presence.
- **Story 2: Theme Layout Selection** – As an instructor, I want to select structured layout frameworks, choose theme accents, and append logos so my landing page looks unique without manual code adjustments.
- **Story 3: Course Architecture** – As an instructor, I want to form courses, split them into topical chapters, and upload video content so students see structured data maps.

Epic 2: The Student Experience

- **Story 4: Course Exploration** – As a student, I want to navigate directly to an instructor's distinct subdomain to view pricing, outlines, and curriculum.
- **Story 5: Adaptive Playback Engine** – As an Egyptian student running on restricted mobile data caps, I want an adaptive player that adjusts streaming rates down to 360p automatically to avoid fast data depletion.

Epic 3: Localized Checkout & Access Control

- **Story 6: Mobile Wallet Manual Rails** – As a student without credit cards, I want to choose Vodafone Cash or InstaPay at checkout, see the tutor's credentials, and upload a payment receipt screenshot.
- **Story 7: Dashboard Management** – As an instructor, I want to analyze receipt uploads via my dashboard and toggle a single "Approve" button to securely activate access tokens.

3. Structural Database Model (PostgreSQL)

```
-- Setup Enums for Quota Matrix & Permissions
CREATE TYPE subscription_tier AS ENUM ('basic', 'pro', 'academy');
CREATE TYPE user_role AS ENUM ('student', 'teacher', 'organization_admin');

-- Top Level Groupings for Schools/Academies
CREATE TABLE organizations (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name VARCHAR(255) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Core Tenant Mapping with Quota Fields
CREATE TABLE tenants (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    organization_id UUID REFERENCES organizations(id) ON DELETE CASCADE,
    subdomain VARCHAR(100) UNIQUE NOT NULL,
    custom_domain VARCHAR(255) UNIQUE NULL,
    tier subscription_tier DEFAULT 'basic',
    theme_settings JSONB DEFAULT '{"template": "default", "primary_color":
"#000000"}'::jsonb,
    storage_used_bytes BIGINT DEFAULT 0,
    active_students_count INT DEFAULT 0,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Profiles Table Associated with Supabase Auth
CREATE TABLE profiles (
    id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
    tenant_id UUID REFERENCES tenants(id) ON DELETE CASCADE,
    full_name VARCHAR(255) NOT NULL,
    role user_role DEFAULT 'student',
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Course Structure Metadata
CREATE TABLE courses (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id UUID REFERENCES tenants(id) ON DELETE CASCADE,
    title VARCHAR(255) NOT NULL,
    price_egp NUMERIC(10, 2) NOT NULL DEFAULT 0.00,
    is_published BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);

-- Lecture Assets Managed via Cloudflare Identifiers
CREATE TABLE lectures (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    course_id UUID REFERENCES courses(id) ON DELETE CASCADE,
    title VARCHAR(255) NOT NULL,
```

```
    cloudflare_stream_id VARCHAR(255) NOT NULL,
    file_size_bytes BIGINT NOT NULL,
    sort_order INT DEFAULT 0
);

-- Manual and Gateway Enrollment Processing
CREATE TABLE enrollments (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    course_id UUID REFERENCES courses(id) ON DELETE CASCADE,
    student_id UUID REFERENCES profiles(id) ON DELETE CASCADE,
    payment_status VARCHAR(50) DEFAULT 'pending',
    payment_receipt_url TEXT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),
    UNIQUE(course_id, student_id)
);
```

4. Core Next.js Multi-Tenant Middleware Routing

```
import { NextResponse } from 'next/server';

export function middleware(request) {
  const url = request.nextUrl.clone();
  const hostname = request.headers.get('host');

  // Skip asset folders, metrics, or native endpoints
  if (
    url.pathname.startsWith('/_next') ||
    url.pathname.startsWith('/api') ||
    url.pathname.startsWith('/static') ||
    url.pathname.includes('.')
  ) {
    return NextResponse.next();
  }

  // Parse raw hostname strings
  const currentHost = process.env.NODE_ENV === 'production'
    ? hostname.replace(`.yourplatform.com`, '')
    : hostname.replace(`.localhost:3000`, '');

  // Safeguard default landing page endpoints
  if (currentHost === hostname || currentHost === 'www' || currentHost === 'app') {
    return NextResponse.next();
  }

  // Intercept and route towards multi-tenant dynamic sub-folders
  url.pathname = `/tenants/${currentHost}${url.pathname}`;
  return NextResponse.rewrite(url);
}
```

5. Scaling and Quota Guardrails Matrix

To safely scale into multi-tenant configurations without incurring direct variable overhead loss, the enforcement metrics are tracked linearly across database operations:

- **Basic Plan (Free / 5% Fee):** 3 GB Max Storage quota, up to 50 active students. Enforces manual payment approvals. Perfect for market validation in Egypt.
- **Pro Plan (250 EGP / mo):** 15 GB Active Storage, 500 maximum active student enrollments. Toggles customized interface controls.
- **Academy Plan (750 EGP / mo):** 100 GB Core Storage, unrestricted student registration limits, and multi-instructor access profiles.